

● ● ● AI Linux Command Assistant Terminal

```
$ ollama run llama3

> How do I monitor CPU usage?

top -bn1 | grep 'Cpu(s)'
```

AI Linux Command Assistant

Intelligent Terminal · Powered by Ollama LLM

User Documentation — Version 1.0.0
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Application	AI Linux Command Assistant
Version	1.0.0
Platform	Windows / Linux / macOS
Language	Python 3.10+
AI Engine	Ollama (Local LLM — No internet required)
GUI Framework	PyQt5
Database	SQLite3

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Tool	AI Linux Command Assistant v1.0.0

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1. Introduction

1.1 What is AI Linux Command Assistant?

AI Linux Command Assistant is a professional desktop application that combines the power of a local Large Language Model (LLM) with a fully functional Linux terminal interface. It allows users to ask natural language questions about Linux commands and receive accurate, context-aware answers — all running locally on your machine with no internet required.

Whether you are a system administrator, DevOps engineer, developer, or student learning Linux, this tool dramatically accelerates your workflow by eliminating the need to manually search documentation or memorize complex command syntax.

1.2 Key Benefits

- ✓ Ask any Linux question in plain English and get instant answers
- ✓ Execute commands directly inside the app without switching windows
- ✓ Save your favorite commands in a searchable personal database
- ✓ All AI processing runs locally — your data never leaves your machine
- ✓ Streaming AI responses appear in real-time as they are generated
- ✓ Full command history with one-click reuse
- ✓ Built-in safety checks to prevent accidental destructive commands
- ✓ Professional dark theme UI with adjustable font sizes
- ✓ Support for multiple Ollama AI models
- ✓ Export and import your favorite commands as JSON

1.3 Technology Stack

Component	Technology	Purpose
GUI Framework	PyQt5	Professional desktop interface
AI Engine	Ollama	Local LLM inference
AI Models	LLaMA 3 / Mistral / CodeLlama	Natural language processing
Database	SQLite3	Favorites storage
Terminal	subprocess	Linux command execution
Syntax Highlight	QSyntaxHighlighter	Code colorization
Packaging	PyInstaller	EXE build

2. System Requirements

2.1 Hardware Requirements

Component	Minimum	Recommended
CPU	Intel Core i5	Intel Core i7 / AMD Ryzen 7
RAM	8 GB	16 GB or more
Storage	10 GB free space	20 GB free space
GPU	Not required	NVIDIA GPU (speeds up LLM)
Display	1280 × 720	1920 × 1080 or higher

2.2 Software Requirements

Software	Version	Notes
Python	3.10 +	Must be installed before running
Ollama	Latest	Free download at ollama.com
PyQt5	5.15 +	pip install PyQt5
ollama (Python)	Latest	pip install ollama
reportlab	Latest	pip install reportlab (for docs)
PyInstaller	Latest	pip install pyinstaller (for EXE)
OS	Any	Windows 10+, Ubuntu 20+, macOS 12+

3. Installation Guide

3.1 Step 1 — Install Python

Download and install Python 3.10 or higher from the official website:

```
https://www.python.org/downloads/
```

■ During installation on Windows, check 'Add Python to PATH' before clicking Install.

Verify the installation by opening a terminal and running:

```
python --version
```

3.2 Step 2 — Install Ollama

Ollama is the local AI engine that powers the assistant. Download and install it from:

```
https://ollama.com
```

After installation, start the Ollama service:

```
ollama serve
```

■ Tip: On Windows, Ollama runs automatically as a system tray app after installation.

3.3 Step 3 — Install Python Dependencies

Open a terminal (or PyCharm terminal) and run:

```
pip install PyQt5 ollama reportlab pyinstaller
```

3.4 Step 4 — Download an AI Model

Pull your preferred AI model. LLaMA 3 is recommended for best results:

```
ollama pull llama3

# Other available models:
ollama pull mistral
ollama pull codellama
ollama pull gemma2
ollama pull phi3
```

■ The first download may take several minutes depending on your internet speed. Model sizes range from 2 GB to 7 GB.

3.5 Step 5 — Run the Application

Navigate to your project folder and run:

```
cd AI_command_assistant  
python ai_linux_assistant.py
```

Or if you have the EXE build:

```
dist\AI_Linux_Assistant.exe
```

■ Tip: Make sure Ollama is running before starting the application, otherwise AI responses will fail.

4. Application Layout

The application is divided into four main areas:

TOOLBAR	Model selector · Font controls · Clear · Save · About
LEFT PANEL	AI Chat display · Input box · Action buttons
RIGHT TOP	Terminal output · Direct command input · Run/Clear
RIGHT BOTTOM	Favorites tab + History tab (tabbed interface)
STATUS BAR	Current status · Active model · Last action

4.1 Toolbar

Element	Description
App Title	Displays application name and version
Model Selector	Dropdown to switch between Ollama AI models
A– Button	Decrease font size by 1 point
Font Size Label	Shows current font size in points
A+ Button	Increase font size by 1 point
■ Reset Button	Reset font size to default (14pt)
■ Clear Chat	Clear the AI conversation history
■ Save Chat	Export the chat log to a text file
■ About	Show application version information

4.2 AI Chat Panel (Left)

The left panel is the main conversation area where you interact with the AI assistant. Messages are displayed as colored chat bubbles — blue for your messages and green for AI responses. Code blocks are automatically highlighted and formatted.

Element	Description
Chat Display	Shows full conversation with formatted bubbles
Thinking Indicator	Shows '■ AI is thinking...' during processing
Input Box	Type your question or command here
■ Send Button	Send your message to the AI
■ Run Last Command	Execute the last command suggested by AI

■ Copy Command	Copy the last AI command to clipboard
■ Save Command	Save last AI command to favorites
✕ Clear Chat	Clear the conversation history

4.3 Terminal Output Panel (Right Top)

The terminal panel displays real-time output from executed commands. Output is color-coded: green for normal output, red for errors, and blue for command headers.

Element	Description
Output Display	Shows stdout and stderr from commands
Command Input	Type Linux commands to run directly
■ Run Button	Execute the typed command
✕ Clear Button	Clear the terminal output
Exit Code	Shows return code and execution time after run

4.4 Favorites & History Tabs (Right Bottom)

The bottom-right section contains two tabs: Favorites and History. Switch between them by clicking the tab labels.

5. Using the AI Assistant

5.1 Asking Questions

Simply type your question in the input box at the bottom of the chat panel and press Enter or click Send. The AI will respond with a detailed explanation and the exact command to use.

Example questions you can ask:

- "How do I check disk usage on my system?"
- "Show me how to monitor CPU usage in real time"
- "How do I find all files larger than 100MB?"
- "What command shows all running processes?"
- "How do I check which ports are open?"
- "How do I restart the nginx service?"
- "Show me how to check network traffic"
- "How do I compress a folder into a tar.gz file?"
- "How do I check system memory usage?"
- "How do I view the last 50 lines of a log file?"

■ Tip: The AI remembers your last 12 messages, so you can ask follow-up questions naturally.

5.2 Running Commands

There are three ways to run commands in the application:

Method	How to Use
■ Run Last Command	Click this button to run the last command the AI suggested
Direct Terminal Input	Type any command in the terminal input box and press Enter or click Run
From Favorites	Double-click any item in the Favorites tab to run it instantly
From History	Double-click any item in the History tab to load it into the command input

■ **Safety Warning:** The application automatically blocks known destructive commands such as 'rm -rf /', 'mkfs', and 'dd if=' to protect your system.

5.3 Switching AI Models

Use the Model dropdown in the toolbar to switch between different Ollama models at any time during a session.

Model	Best For	Size
llama3	General Linux help	~4.7 GB
llama3.2	Faster responses	~2.0 GB
mistral	Detailed explanations	~4.1 GB
codellama	Command & script help	~3.8 GB

gemma2	Concise answers	~5.0 GB
phi3	Lightweight / fast	~2.3 GB

■ Tip: Use 'codellama' for script writing and 'llama3' for general system administration questions.

6. Favorites Database

The Favorites system lets you build a personal database of your most-used Linux commands, organized by category, with custom names and descriptions. Data is stored in a local SQLite database file (favorites.db).

6.1 Adding Favorites

- 1. Click the '■ Add New' button in the Favorites tab
- 2. Fill in the Name field (required)
- 3. Enter the Command (required)
- 4. Select a Category from the dropdown
- 5. Add an optional Description
- 6. Click Save

■ Tip: You can also click '■ Save Command' in the chat panel to save the last AI-suggested command directly to favorites.

Available Categories:

General	Network	Files	Process
System	Disk	Testing	Security
Docker	Custom		

6.2 Searching and Filtering

The Favorites tab includes a live search bar that instantly filters your favorites as you type. You can also filter by category using the dropdown next to the search bar.

Feature	How It Works
Live Search	Type in the search box — results update instantly
Search Fields	Searches name, command, description, and category
Category Filter	Select a category to show only those commands
Clear Filter	Select 'All Categories' to show everything

6.3 Editing and Deleting

Select any favorite in the list and use the action buttons:

Button	Action
✎ Edit	Opens the edit dialog with current values pre-filled
■ Delete	Shows confirmation dialog before deleting
■ Run	Runs the selected favorite command immediately
Double-click	Also runs the command immediately

6.4 Import and Export

You can back up and share your favorites using JSON files:

```
# Export format (favorites_export.json):  
[  
  {  
    "name": "Check Disk Usage",  
    "command": "df -h",  
    "category": "Disk",  
    "description": "Shows disk usage in human-readable format",  
    "created_at": "2025-01-01 12:00:00"  
  }  
]
```

7. Command History

Every command you run is automatically saved to the History tab and stored in the `command_history.json` file. The history persists between sessions.

Feature	Description
Auto-save	Commands are saved automatically on execution
Persistent	History survives app restarts
Double-click to load	Loads command into the terminal input box
■ Use Button	Loads selected command into terminal input
■ Save to Favorites	Promotes a history item to your favorites
✕ Clear History	Removes all history entries
Max entries	100 most recent commands are kept

8. Font Size and Display

The font size is fully adjustable and affects all panels simultaneously — the chat panel, terminal, input boxes, and history list. Your preferred size is saved to `settings.json` and restored automatically on next launch.

Control	Action
A+ Button	Increase font size by 1 point
A– Button	Decrease font size by 1 point
■ Reset	Reset to default size (14 pt)
Ctrl + = or +	Increase font size (keyboard shortcut)
Ctrl + –	Decrease font size (keyboard shortcut)
Ctrl + 0	Reset to default size (keyboard shortcut)
Range	10 pt minimum — 26 pt maximum
Auto-save	Font size saved to <code>settings.json</code> on change

9. Keyboard Shortcuts

Shortcut	Action
Enter	Send message / Run command
Ctrl + =	Increase font size
Ctrl + -	Decrease font size
Ctrl + 0	Reset font size to default
F11	Toggle fullscreen mode
Ctrl + C	Copy selected text
Ctrl + A	Select all text in input

10. Supported AI Models

The assistant supports all models available through Ollama. Below are the recommended models for this application:

Model	Speed	Quality	Size	Best Use Case
llama3	Medium	Excellent	4.7 GB	General Linux help
llama3.2	Fast	Very Good	2.0 GB	Quick answers
mistral	Medium	Excellent	4.1 GB	Detailed guides
codellama	Medium	Excellent	3.8 GB	Scripts & code
gemma2	Medium	Very Good	5.0 GB	Concise answers
phi3	Very Fast	Good	2.3 GB	Lightweight use

To add a new model, run in your terminal:

```
ollama pull <model_name>
# Example:
ollama pull codellama
```

■ Tip: Browse all available models at <https://ollama.com/library>

11. Troubleshooting

■ AI returns error: 'Ollama not running'

Cause: Ollama service is not started

Fix:

```
ollama serve
```

■ Model not found error

Cause: Model was not downloaded

Fix:

```
ollama pull llama3
```

■ App window is blank or crashes

Cause: PyQt5 not installed or wrong version

Fix:

```
pip install --upgrade PyQt5
```

■ Font is very small

Cause: Default font size too small for your display

Fix:

```
Use A+ button or Ctrl + = to increase font size
```

■ Command output shows permission denied

Cause: Command requires elevated privileges

Fix:

```
Prefix with sudo (e.g. sudo df -h)
```

■ Favorites not saving

Cause: favorites.db file permissions issue

Fix:

```
Run app from a folder you have write access to
```


■ EXE build crashes on launch

Cause: Missing data files not bundled

Fix:

```
Use --add-data flag in PyInstaller command
```

■ AI responses are slow

Cause: Model is large or no GPU available

Fix:

```
Switch to phi3 or llama3.2 for faster responses
```

12. Frequently Asked Questions

Q: Does this app send my data to the internet?

A: No. All AI processing is done locally using Ollama. Your commands and conversations never leave your machine.

Q: Can I use this app on Windows?

A: Yes. The app runs on Windows 10+, Linux, and macOS. On Windows, commands are executed via the Windows shell (cmd.exe / PowerShell).

Q: How do I update the AI model?

A: Run 'ollama pull ' in your terminal to get the latest version of any model.

Q: Can I add my own custom categories to favorites?

A: Yes. The 'Custom' category is available for this purpose. Additional categories can be added by editing the CATEGORIES list in the source code.

Q: How many favorites can I save?

A: There is no limit. The SQLite database can store thousands of favorites with no performance impact.

Q: Can I share my favorites with a teammate?

A: Yes. Use the Export JSON button to save your favorites to a file, then share that file. Your teammate can import it using the Import JSON button.

Q: What happens if I type a wrong command?

A: The terminal will show the error output in red. No changes are made to your system until a command actually runs successfully.

Q: Can I run multiple commands at once?

A: Yes. Use shell operators like && or ; to chain commands. Example: `cd /var/log && ls -la`

13. Safety and Security

The application includes built-in safety features to prevent accidental execution of dangerous commands. However, users must exercise caution when running any system command.

Built-in Blocked Commands:

Blocked Pattern	Reason
<code>rm -rf /</code>	Deletes entire root filesystem
<code>mkfs</code>	Formats a disk partition
<code>> /dev/sda</code>	Overwrites raw disk device
<code>dd if=</code>	Raw disk copy — can destroy data

Best Practices:

- Always read and understand a command before running it
- Do not run commands with `sudo` unless you understand the effect
- Test destructive commands on non-production systems first
- Back up important data before running disk or file operations
- Use the AI to explain what a command does before executing it
- Do not share your `favorites.db` if it contains sensitive info

End of Documentation

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